To what extent does enhanced agent differentiation impact the performance of flocking simulations

Zack Younger

Durham University
Department of Computer Science

February 10, 2026

Contents

1 Introduction **2**

2 Literature Analysis **3**

3 Methodology **4**

3.1 Boids 4

3.1.1 Perception 4

3.1.2 Separation 4

3.1.3 Cohesion 5

3.1.4 Alignment 5

3.1.5 Utilising the rules 5

3.2 Agent Differentiation 6

3.2.1 Agent Leadership 6

3.3 Acceleration Structures 6

3.3.1 Spatial Partitioning 6

3.3.2 Octrees 6

3.4 Evaluation 6

3.4.1 Performance 6

3.4.2 Accuracy 6

4 Conclusion **7**

A Appendix A **8**

Bibliography **9**

Chapter 1

Introduction

Chapter 2

Literature Analysis

Chapter 3

Methodology

3.1 Boids

First I will discuss the core Boids algorithm, based on Reynolds' original model [1], that makes up the basis of my work.

We utilise a set B of n agents:

$$B = \{b_0, b_1, \ldots, b_{n-1}\}$$

Where each agent b_i has a position vector $\vec{p}_i$ and a velocity vector $\vec{v}_i$. The agents move in a continuous 3D space according the three primary rules; Separation, Alignment, and Cohesion.

3.1.1 Perception

Importantly, Boid's movement rules are based on local perception. Each agent b_i has a perception radius r, and can only perceive other agents b_j that are within this radius. In the standard definition of Boids[1], Boids can see all around them.

These neighbours are then utilised to calculate the steering forces of the three primary rules.

3.1.2 Separation

The separation rule allows the Boids to avoid collisions and overcrowding. For each agent b_i, we calculate a separation vector $\vec{s}_i$ that steers the agent away from its neighbours. This is done by summing the inverse of the displacement vectors to each neighbour:

$$\vec{s}_i = -\sum_{b_j \in N_i} (\vec{p}_i - \vec{p}_j)$$

3.1.3 Cohesion

The cohesion rule steers the Boids towards the average position of their neighbours. This counteracts the separation rule and helps maintain the flock. The cohesion vector $\vec{c_i}$ is calculated as:

$$\vec{c_i} = \frac{1}{|N_i|} \sum_{b_j \in N_i} \vec{p_j} - \vec{p_i}$$

3.1.4 Alignment

The alignment rule steers the Boids to match the average velocity of their neighbours. This helps the flock move in a unified direction. The alignment vector $\vec{a}_i$ is calculated as:

$$\vec{a}_i = \frac{1}{|N_i|} \sum_{b_j \in N_i} \vec{v}_j - \vec{v}_i$$

3.1.5 Utilising the rules

The final steering vector $\vec{f_i}$ for each agent b_i is a weighted sum of the three rules:

$$\vec{f_i} = w_s \vec{s_i} + w_c \vec{c_i} + w_a \vec{a}_i$$

Where w_s, w_c, and w_a are the weights for separation, cohesion, and alignment respectively. We can tune these weights to adjust the behaviour of the flock. Finally, the agent's velocity and position are updated based on the steering vector:

$$\vec{v}_i = \vec{v}_i + \vec{f_i}$$

3.2 Agent Differentiation

3.2.1 Agent Leadership

3.3 Acceleration Structures

3.3.1 Spatial Partitioning

3.3.2 Octrees

3.4 Evaluation

3.4.1 Performance

The main metric that I will be utilising to compare and evaluate different methodologies will be the time taken.

Acceleration Structure Performance

For the acceleration structures, I will be measuring the time taken to query the neighbours for each agent. This will be done by running the simulation for a fixed number of steps and measuring the average time taken per step for neighbour queries. I will vary the number of agents to effectively compare and contrast the performances under different loads.

3.4.2 Accuracy

Firstly to ensure that I have an effective flocking algorithm, I will be using Polarisation. It is a critical metric for evaluating the collective behaviour of the flock. It measures how aligned the agents are in their movement, with 1 indicating perfect alignment and 0 indicating random movement.

$$\Phi = \frac{1}{N} \left| \sum_{i=1}^{N} \hat{\mathbf{v}}_i \right|$$

This allowws us to utilise Polarization Susceptibility, which is the variance of the polarization over time, used to measure the stability of the flocking behaviour. Susceptibility can be thought as as the responsiveness of the system.

$$\chi = N(\langle \Phi^2 \rangle - \langle \Phi \rangle^2)$$

Chapter 4

Conclusion

Appendix A

Appendix A

Bibliography

[1] Craig W Reynolds. "Flocks, herds and schools: A distributed behavioral model". In: *ACM SIGGRAPH computer graphics* 21.4 (1987), pp. 25–34.